

Data Assimilation in Fluid Dynamics: Evaluation and Analysis

Autor: Tu Nombre

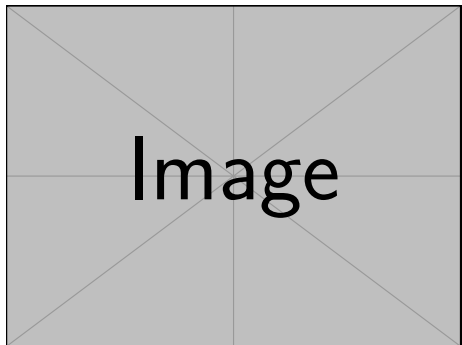
November 20, 2025

Motivation

- CFD simulations are accurate but computationally expensive.
- Low-resolution models are cheaper but less precise.
- Can we use sparse observations to correct coarse simulations?
- Data assimilation (DA) provides a formal solution.

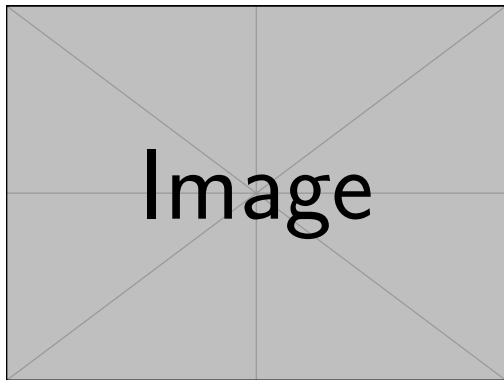
State of the Art

- DA integrates observational data into CFD simulations [5, 2].
- Methods: Kalman filters, variational approaches, neural models like PINNs and FNOs [3, 4].
- Applied to turbulent flows, biomedical modeling, and urban prediction [1].



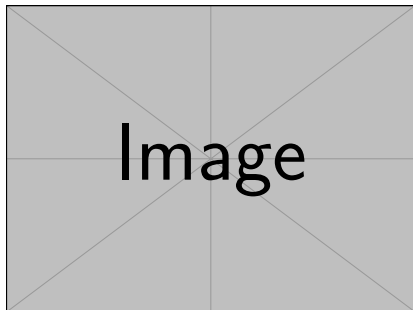
Model Overview

- Base simulation: 2D advection-diffusion on a 25×25 grid.
- High-resolution reference: 800×800 grid (truth field).
- Comparison via RMSE between coarse and true fields.



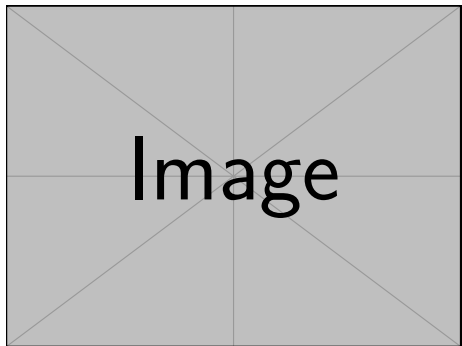
Data Assimilation Methods

- **Nudging**: simple feedback toward observations.
- **KF**: linear filter optimal for Gaussian noise.
- **EnKF**: ensemble-based approximation.
- **4DVar**: optimization-based over time window.



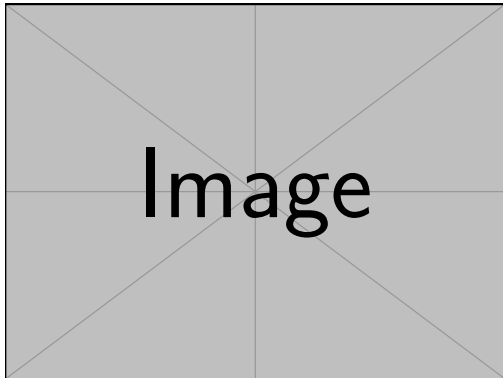
DA Implementation

- Observations added every 70, 150, or 300 timesteps.
- Variables: T , p , τ (alone or combined).
- Sensors: 5, 10, or 20 randomly distributed.
- Output: simulations stored as .pkl and analyzed offline.



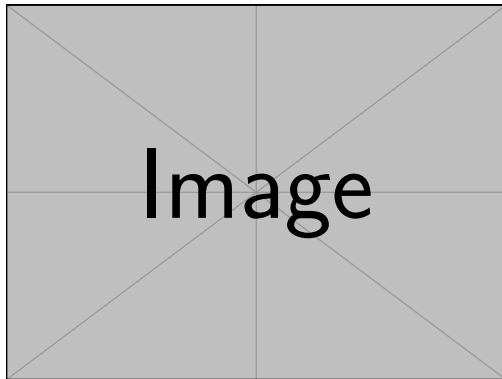
Evaluation Pipeline

- Interpolation of reference to 25×25 grid.
- Global normalization of variables T and p .
- RMSE computed per frame and averaged.
- Best simulations: lowest RMSE and lowest computation time.



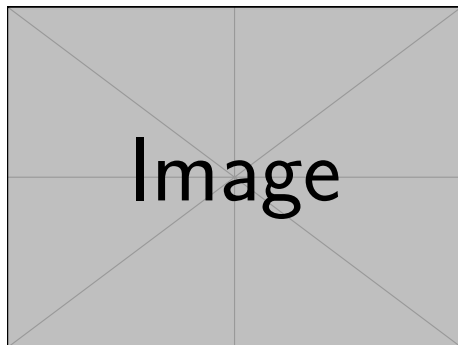
Key Results

- **Best accuracy:** EnKF with full observation set (T, p, τ) and high frequency.
- **Best speed:** Nudging with fewer sensors.
- More sensors $\Rightarrow$ better performance, but diminishing returns.
- Variable combinations improve results significantly.



Conclusions & Next Steps

- DA improves coarse CFD simulations using sparse data.
- Trade-off: accuracy vs computation time.
- Next: explore hybrid neural-DA schemes (PINNs, FNO).
- Real-time integration and 3D generalization as future goals.



References I



J. Bae, Y. Lee, J.H. Rho, G. Kang, and J.J. Kim.

Cfd-quality nowcasting for urban air mobility with a deep learning-based emulator.

Environmental Research Letters, 2025.



X. Chen, G. Wang, and Z. Ye.

Wind tunnel wall interference correction for transonic airfoils with data-reduced ensemble kalman filter.

Physics of Fluids, 2024.



O.L. Cruz-González, V. Deplano, and B. Ghattas.

Enhanced vascular flow simulations in aortic aneurysm via physics-informed neural networks and deep operator networks.

arXiv preprint arXiv:2503.17402, 2025.



Y. Xie, B. Deng, C. Jiang, and C. Lv.

A boundary-assimilation fourier neural operator for predicting initial fields of flow around structures.

Physics of Fluids, 2025.



Y. Zhang, C. Yang, K. Huang, T. Huang, and W. Gui.

Strom: A spatial-temporal reduced-order model for zinc fluidized bed roaster temperature field prediction.

Engineering, 2025.